

Sold Items

112,65 Tsd.

Average Processing Time in hours

78,05

Median Processing Time in hours

53,00

Total Sales

12,25 Mio.

Average Processing Time in days

3,25

Median Processing Time in days

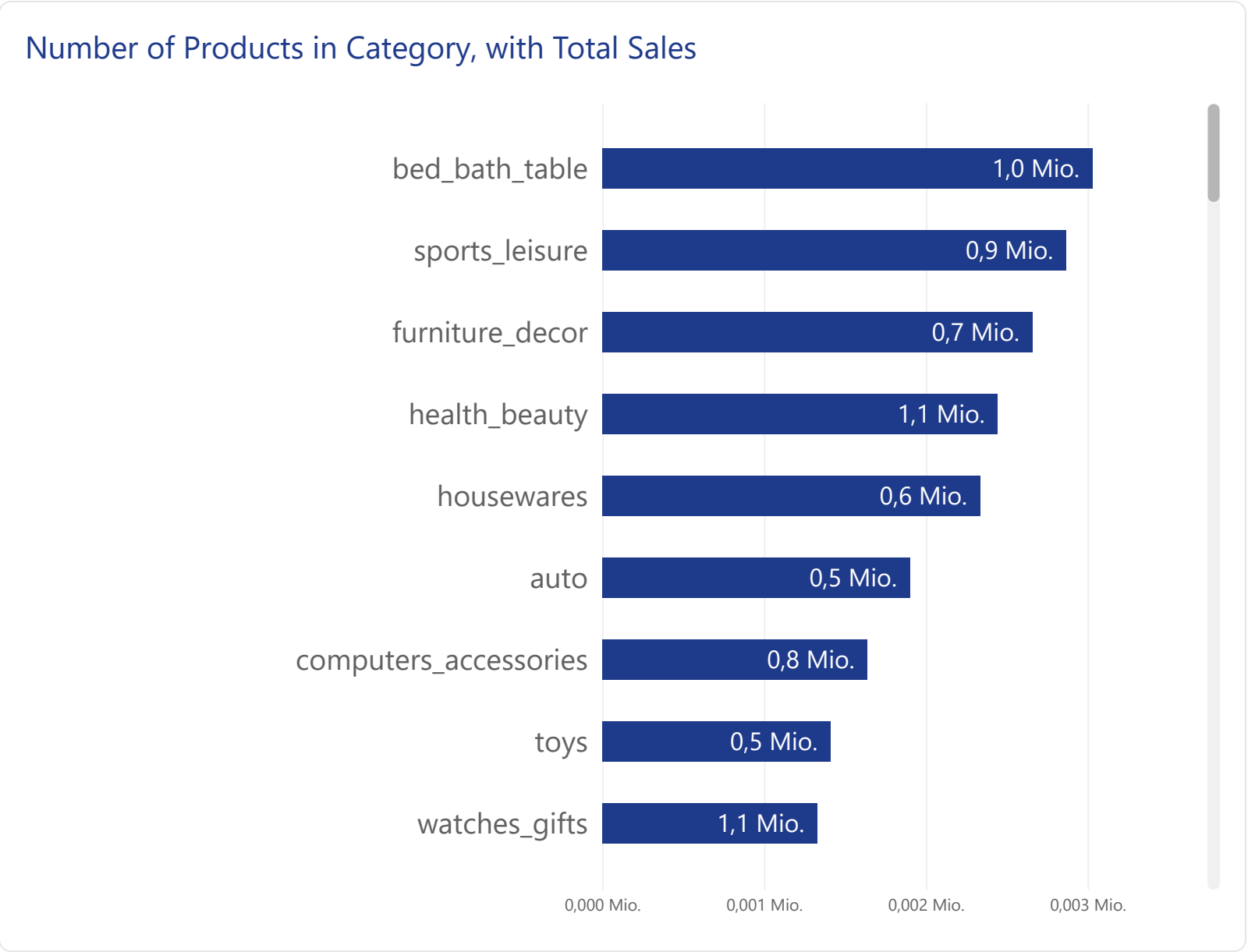
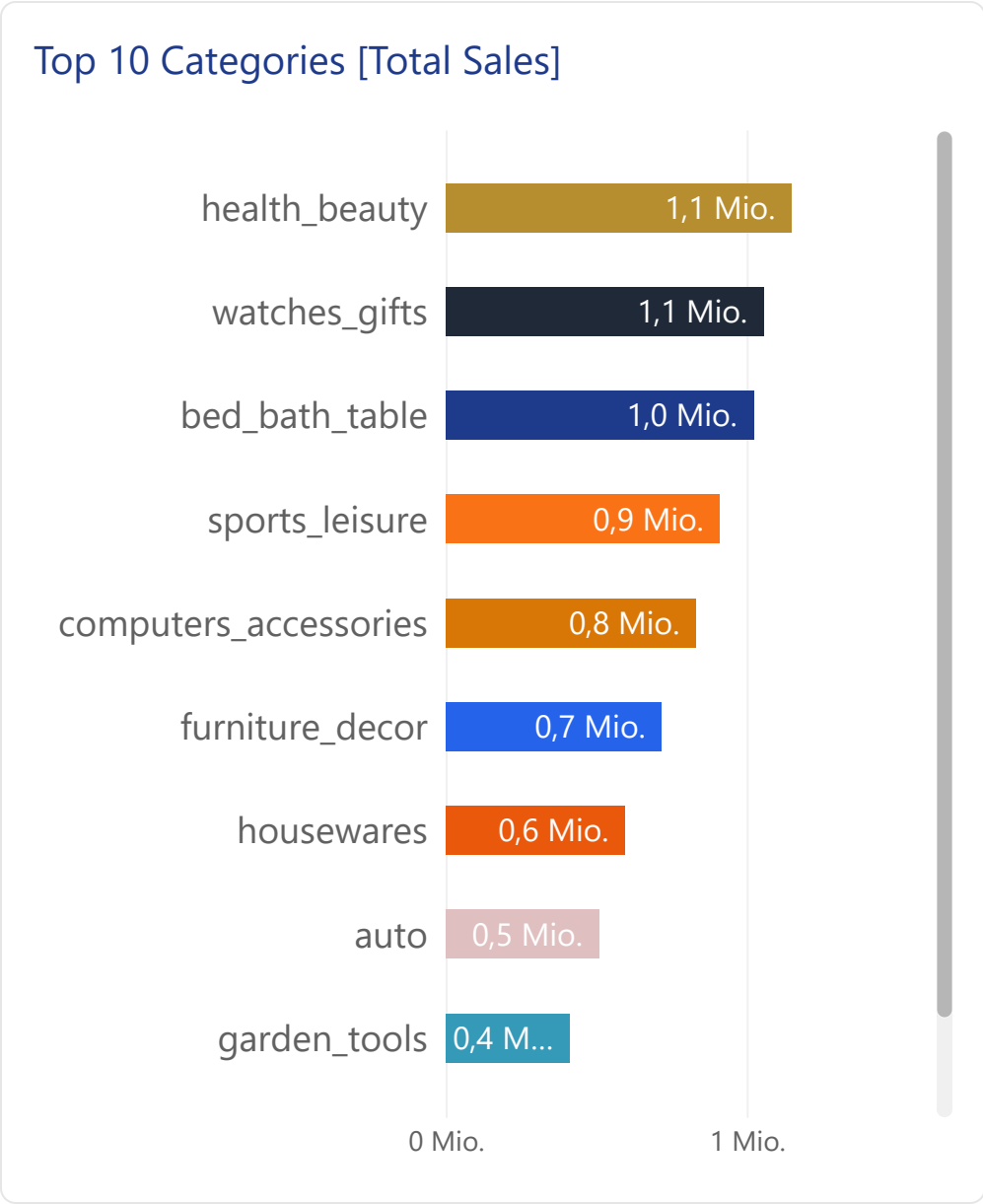
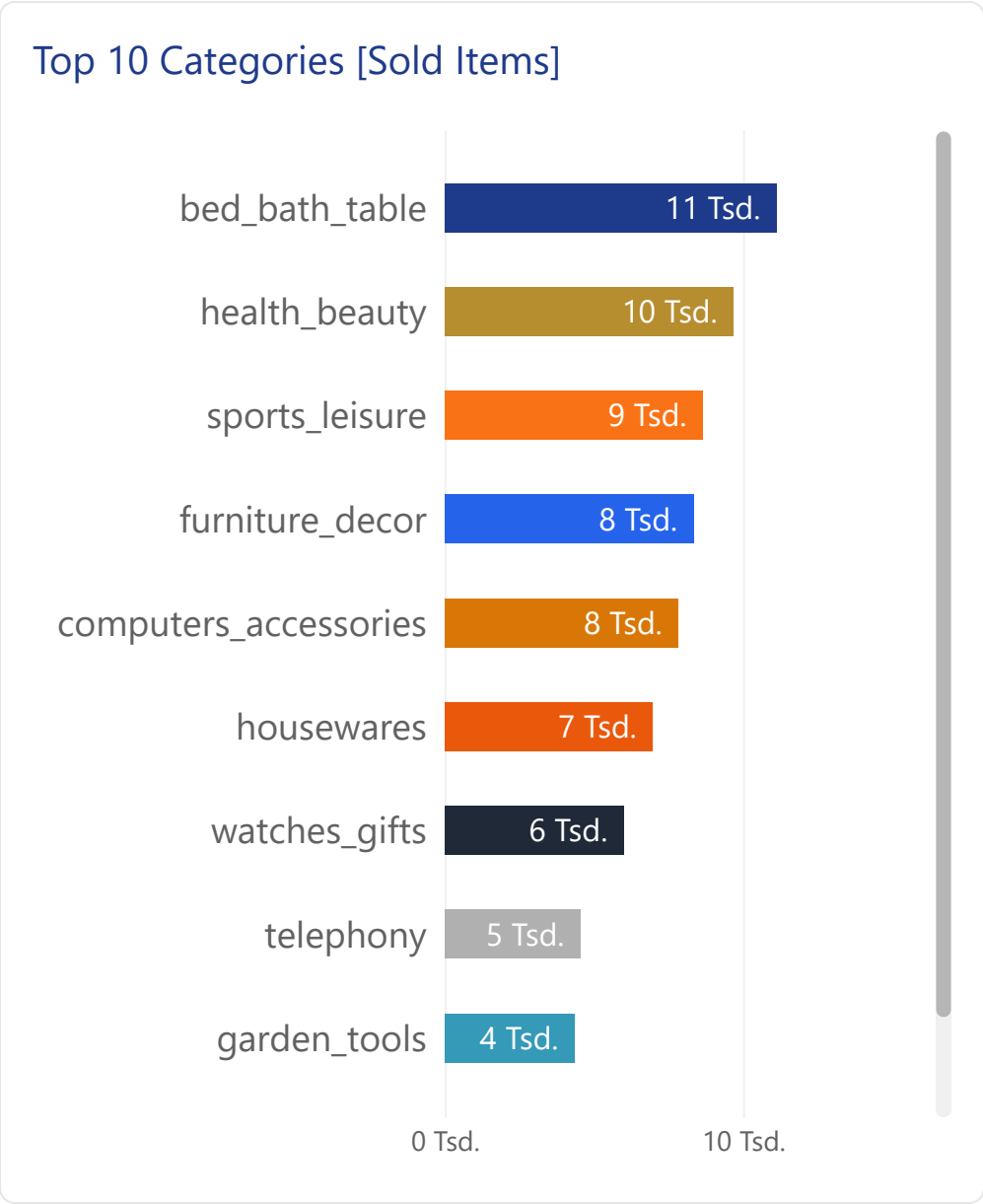
2,21

Year of Purchase

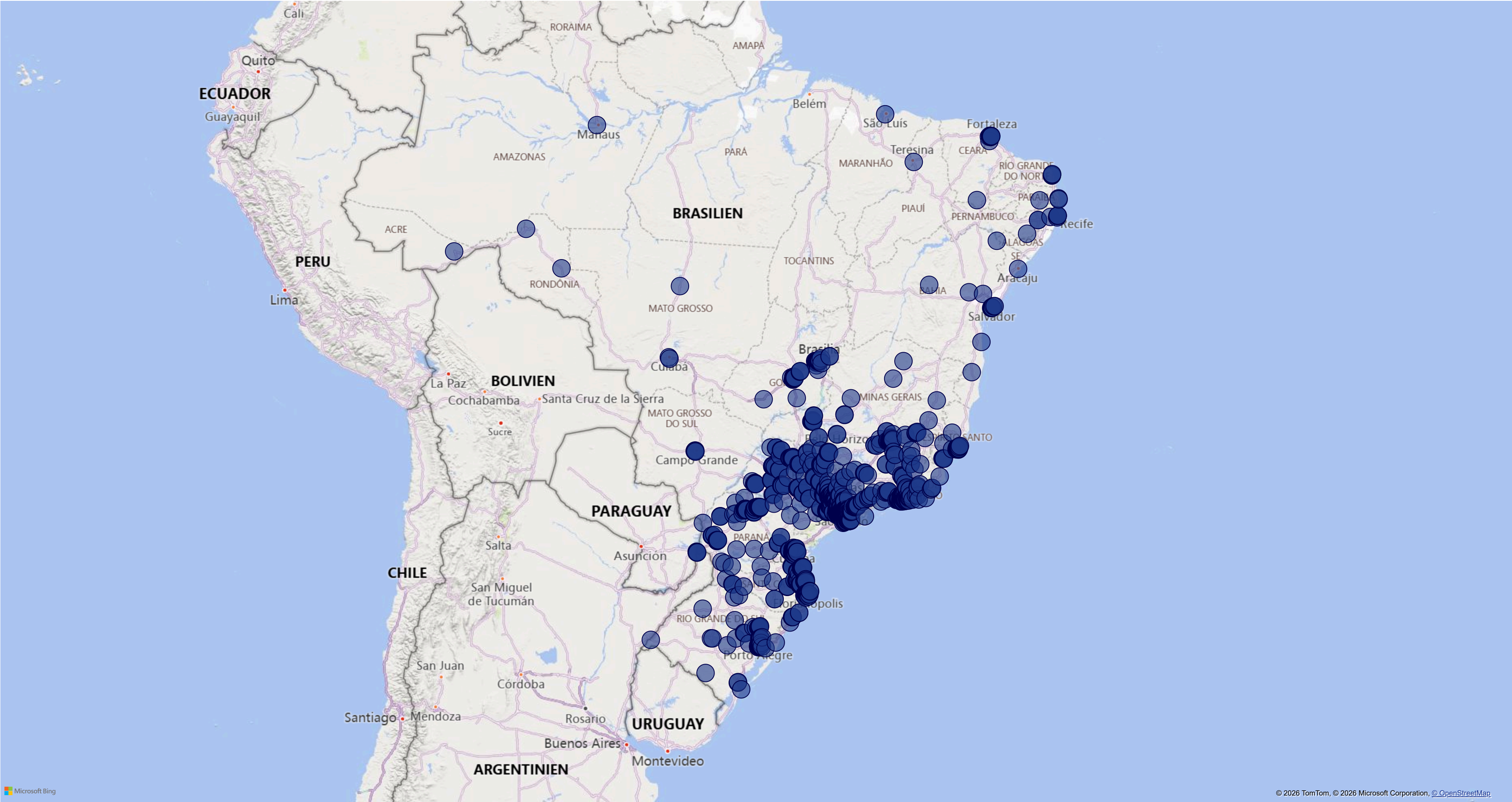
○ 2016

○ 2017

○ 2018

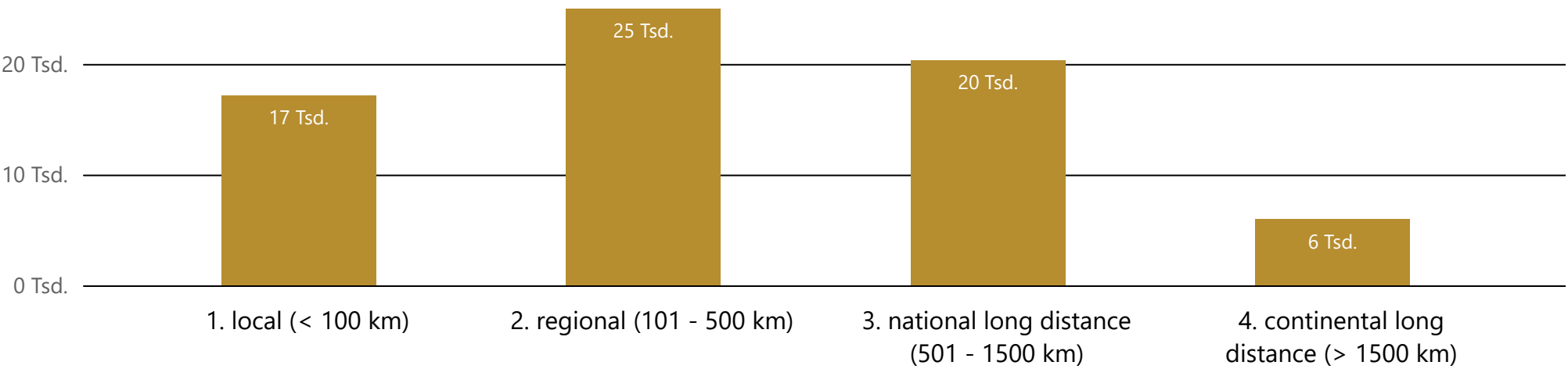


distribution of sellers in brasil



Overview of distances in brasil

sales by distance



average delivery distance in km

564

max delivery distance

3398

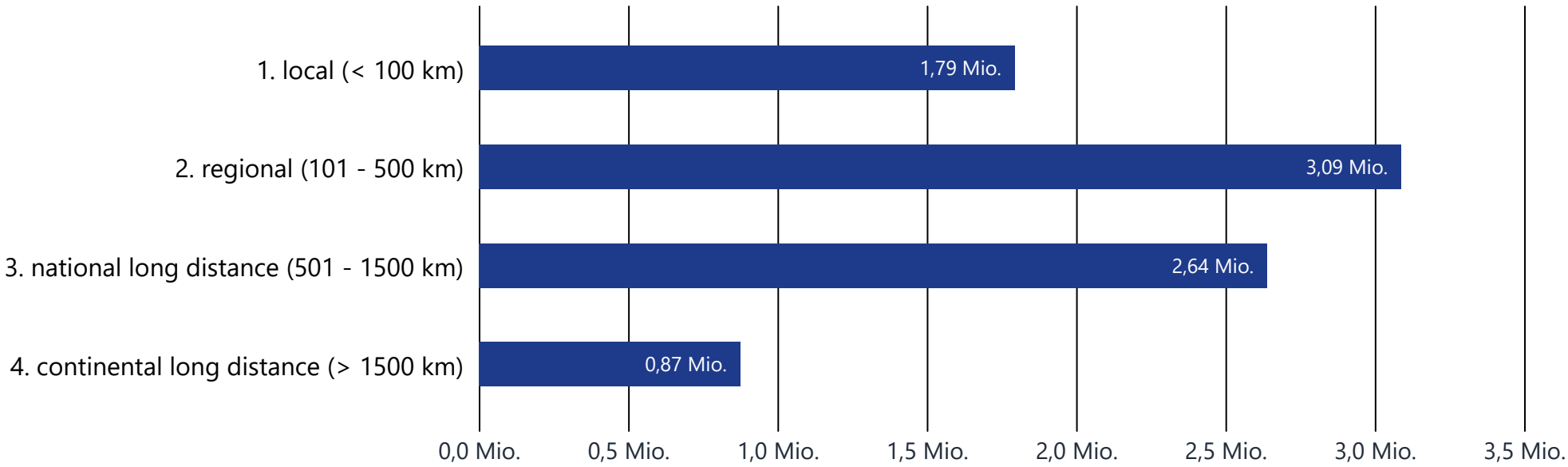
median of delivery distance in km

393

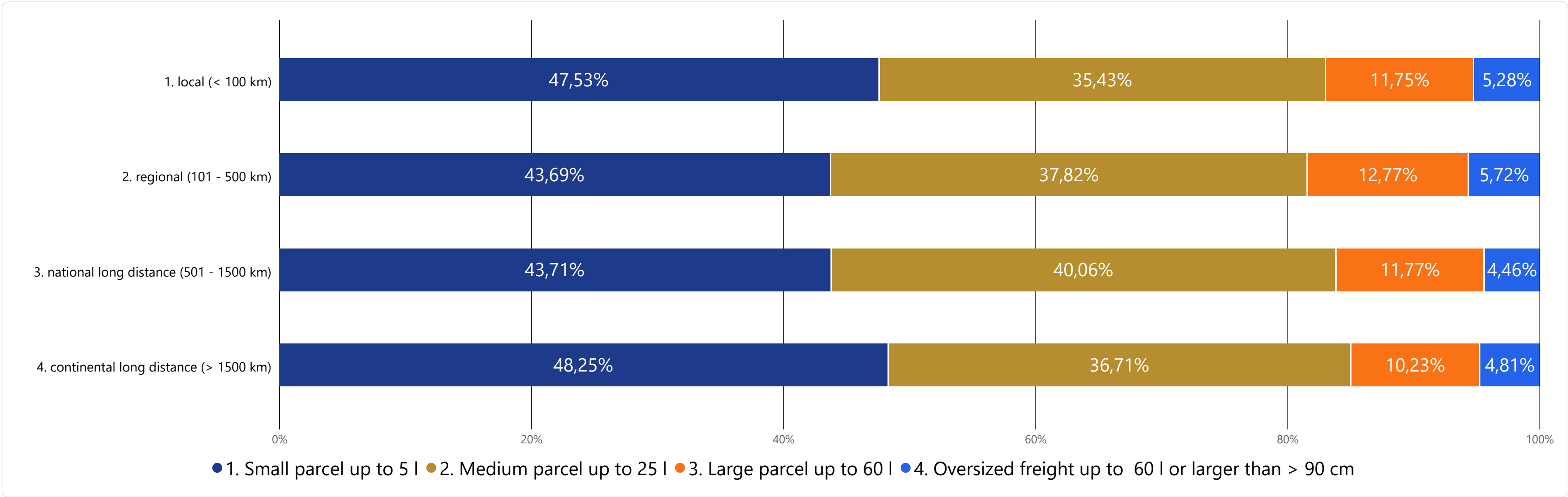
min delivery distance

0,17

total revenue by distance



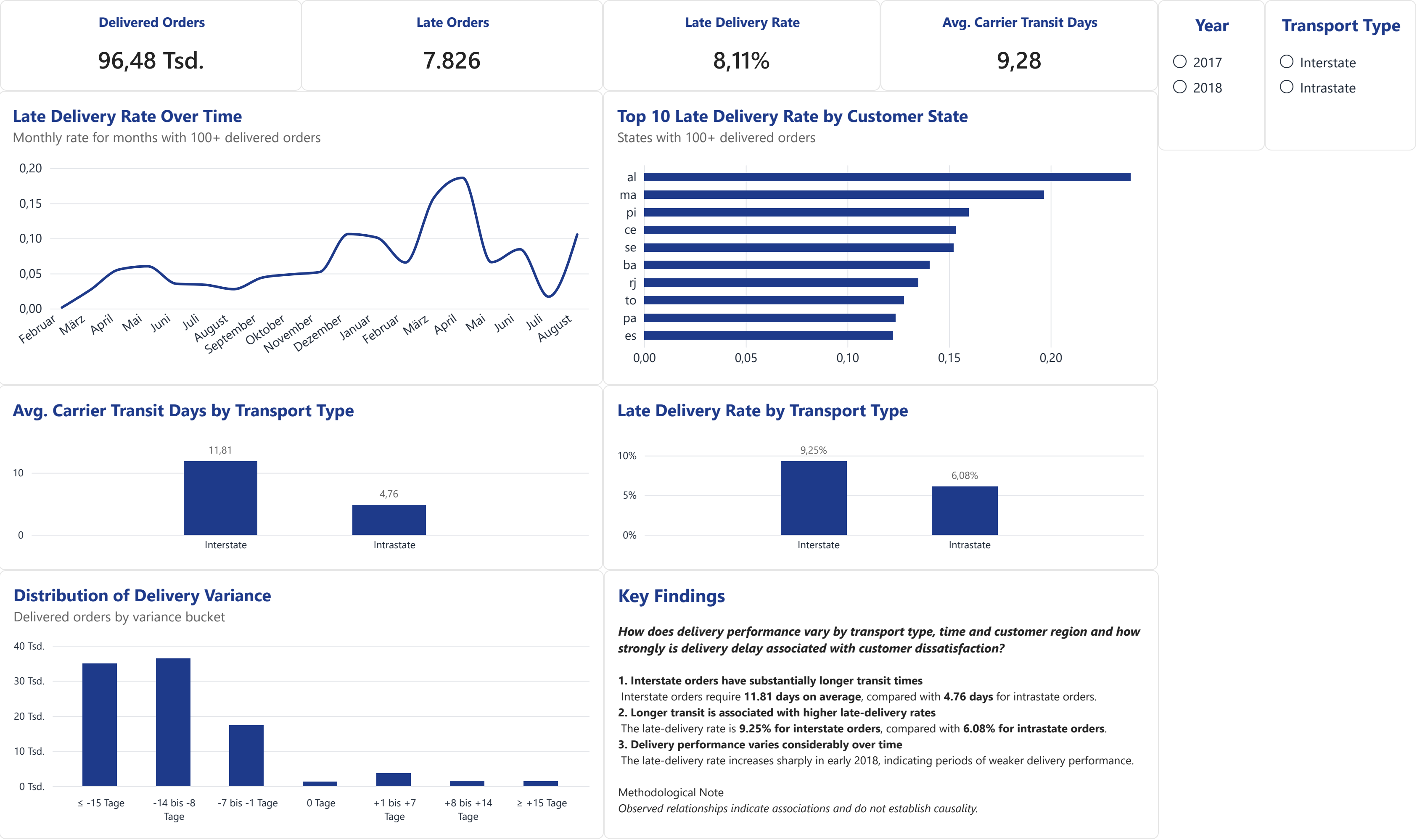
breakdown of parcelsizes by distance



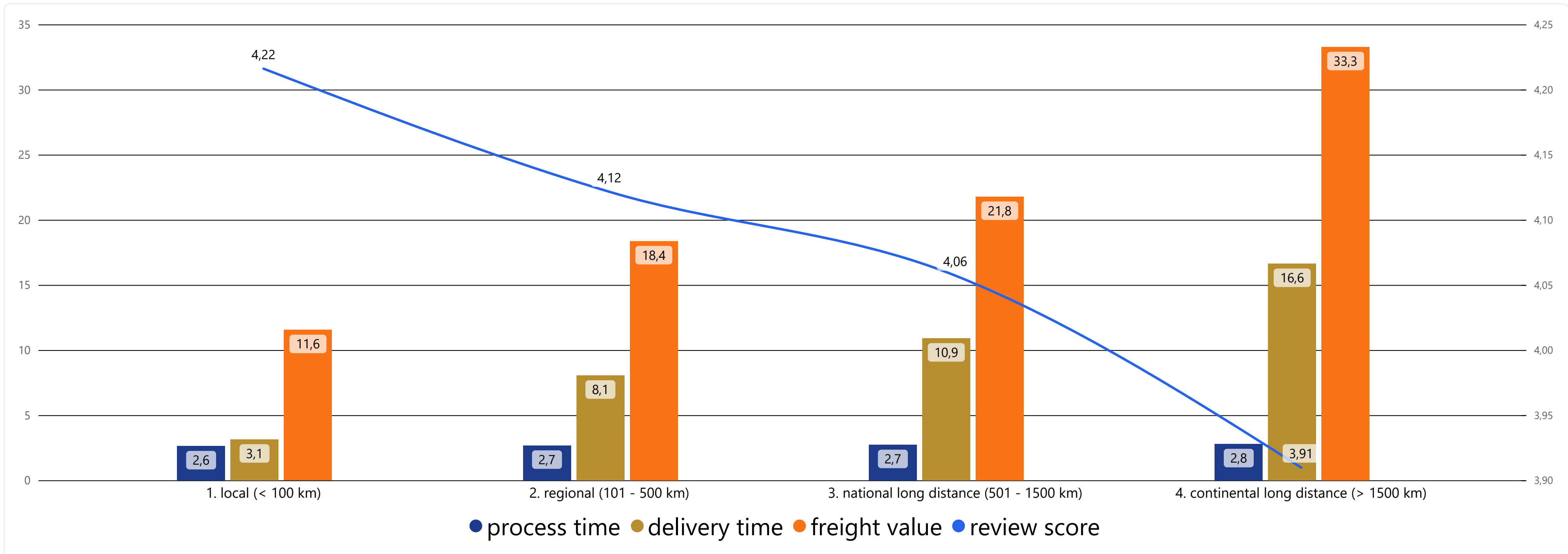
Key findings:

- Small and medium parcels account for the majority of shipments
- Oversized freight volumes reman stable, regardless of distance

Carrier Analysis



impact on delivery time



Key Findings:

- the packaging time remains constant in all four categories of distance. The sellers are working well in this discipline.
- the delivery time rises of course with distance. From 11 to 16 approx in distance categories 3 and 4
- the cost of delivery goes up by almost 50 % in category 3 and 4.
- The review score falls by 0.15 from category 3 to 4 and by 0.10 from category 1 to 2. The features have a small influence on the review score.

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First Order Date per Seller

04.09.2016

Total Sales

12,25 Mio.

Average Processing Time in days

3,25

Median Processing Time in days

2,21

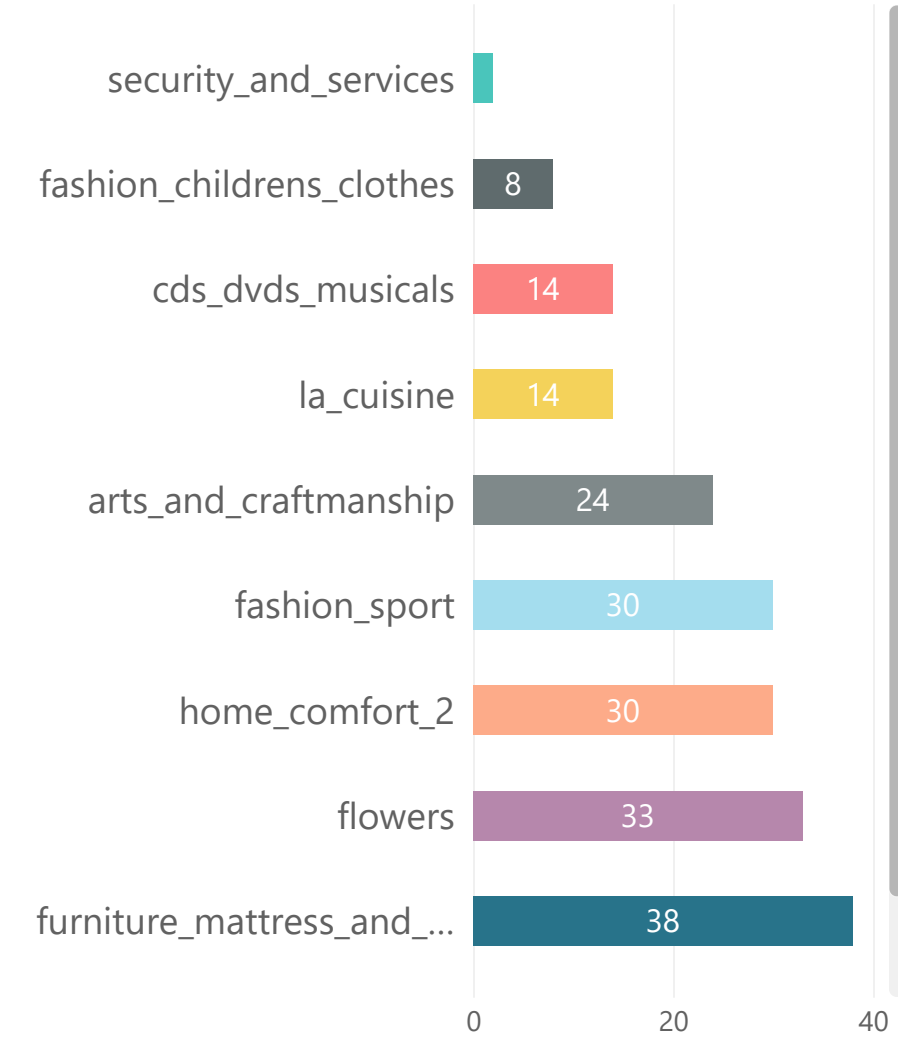
Last Order Date per Seller

03.09.2018

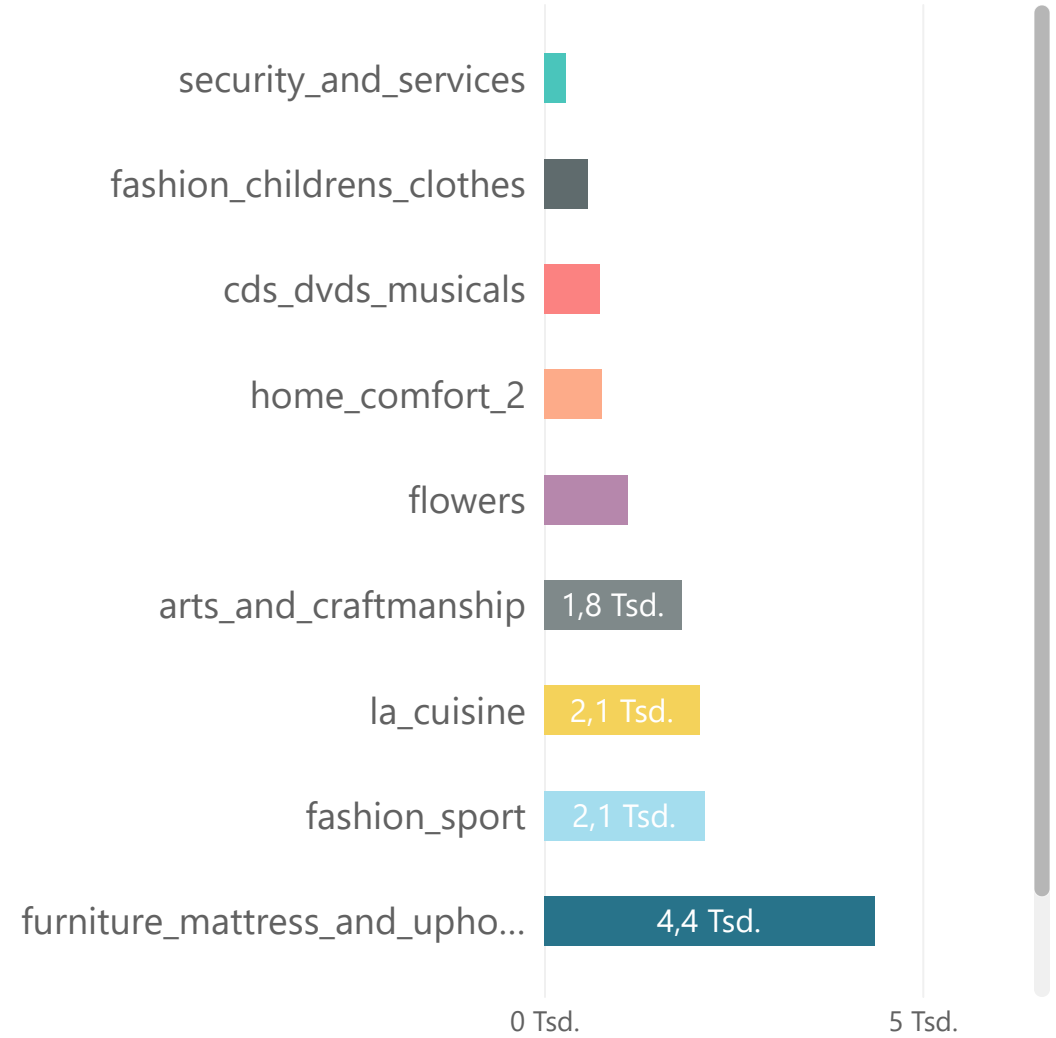
Year of Purchase

- ☐ 2016
- ☐ 2017
- ☐ 2018

Flop 10 Categories [Sold Items]



Flop 10 Categories [Total Sales]



Flop 100 Sellers [Total Sales]

Last Sale at least 1 year ago

Seller	Sold Items	Total Sales	First Order Date	Last Order Date
139157dd4daa45c25b0807ffff348363	1	29,90	31.08.2017	31.08.2017
bd65d163ded559eb90a10cd09fec178f	2	19,80	21.08.2017	21.08.2017
e174dc2de86d60fee178170a154c05e7	1	19,80	17.08.2017	17.08.2017
e76ecc7e9a1a1e3c6df9d2a8261c423d	1	20,00	13.08.2017	13.08.2017
98f52e1020fe788c10c0418de4393fec	1	21,60	09.08.2017	09.08.2017
34675b28b826facfde119785bf287dc1	1	21,99	01.08.2017	01.08.2017
72c73be2b085b9d57650dd53eb2004c9	1	29,40	24.07.2017	24.07.2017
1a6245add4353f18ed55138808906979	1	30,00	22.07.2017	22.07.2017
b026643914f651e8371425e5b8a1e780	1	27,90	07.07.2017	07.07.2017
6614814a00d344b846ae209f95ee7e3f	1	13,00	08.06.2017	08.06.2017
804287717b9156fb7a787acd9af4fac1	1	18,80	07.06.2017	07.06.2017
6c9875b2f94ba781186f0c1aed8d1687	4	25,00	21.05.2017	13.08.2017
fb503a924a0b9db19d83dd0ac6dbef8c	1	13,90	16.05.2017	16.05.2017
34aefe746cd81b7f3b23253ea28bef39	1	8,00	05.05.2017	05.05.2017
cf6f6bc4df3999b9c6440f124fb2f687	1	3,50	01.05.2017	01.05.2017
1c83418358f249c9d6affa52d579db0e	1	24,90	29.04.2017	29.04.2017

Average Processing
Time in hours

78,05

Median Processing
Time in hours

53,00

Average Processing
Time in days

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Median Processing
Time in days

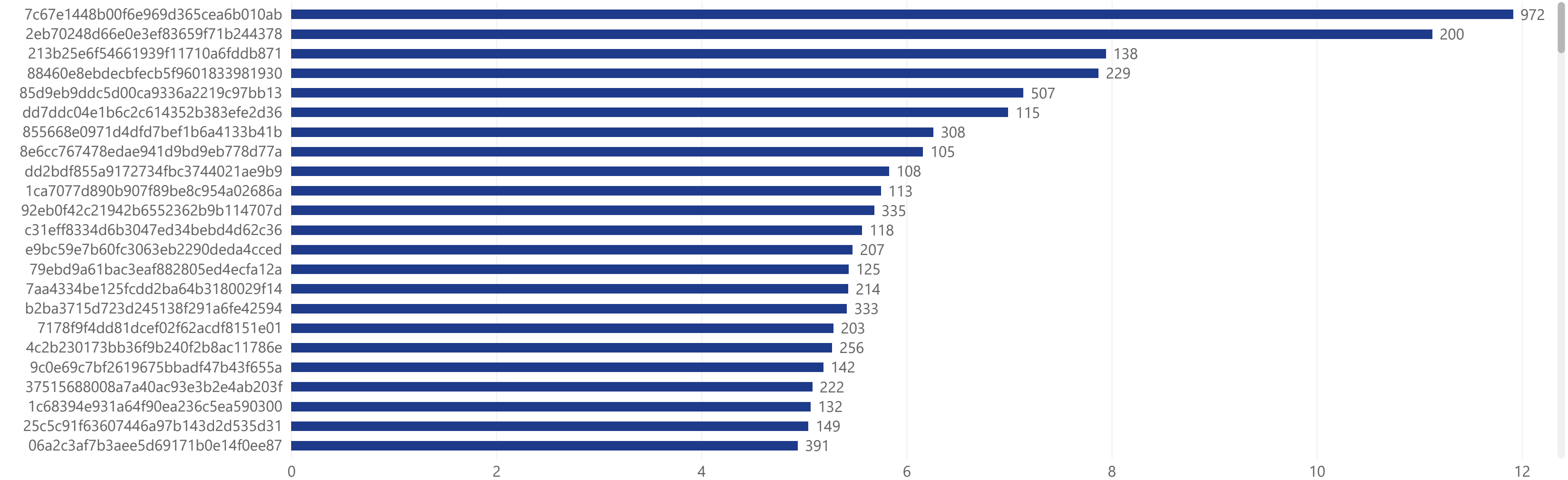
2,21

Average processing times are
significantly higher than the
median.

Individual sellers are bringing the
average down.

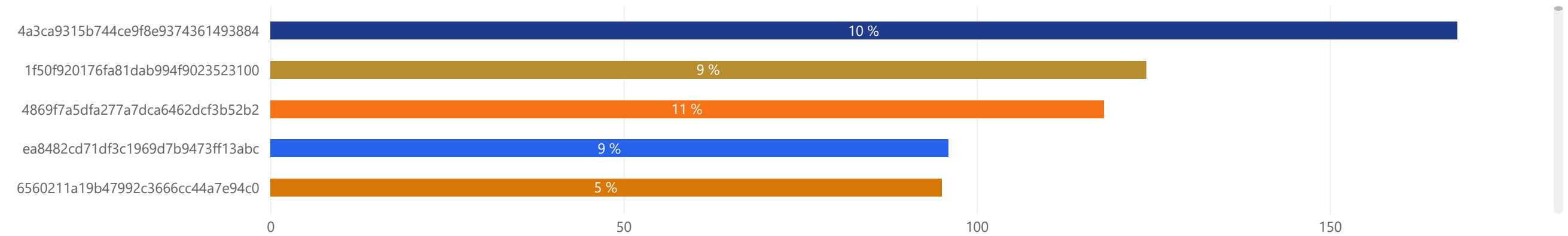
Average Processing Times per Seller in Days, with Number of Orders

Single Seller Orders > 100



Sellers with highest number of delayed Orders, with delayed rate of the Seller

Single Seller Orders



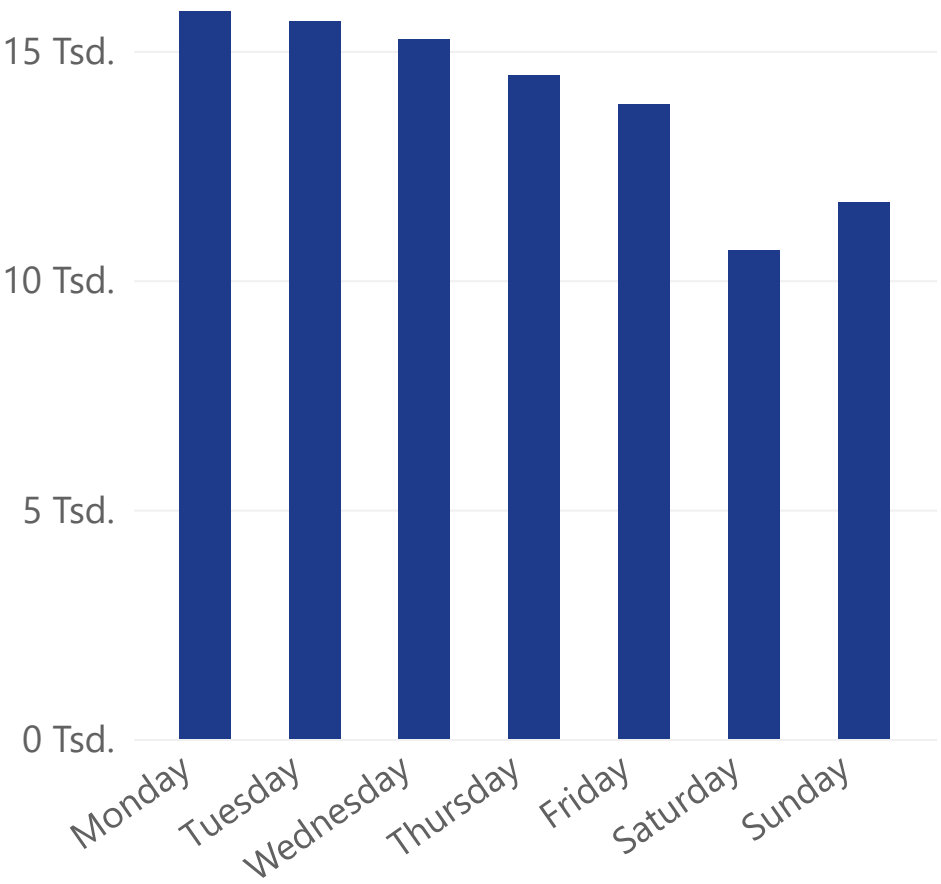
Seller

0015a82c2db000af6aaaf3ae2ecb0532

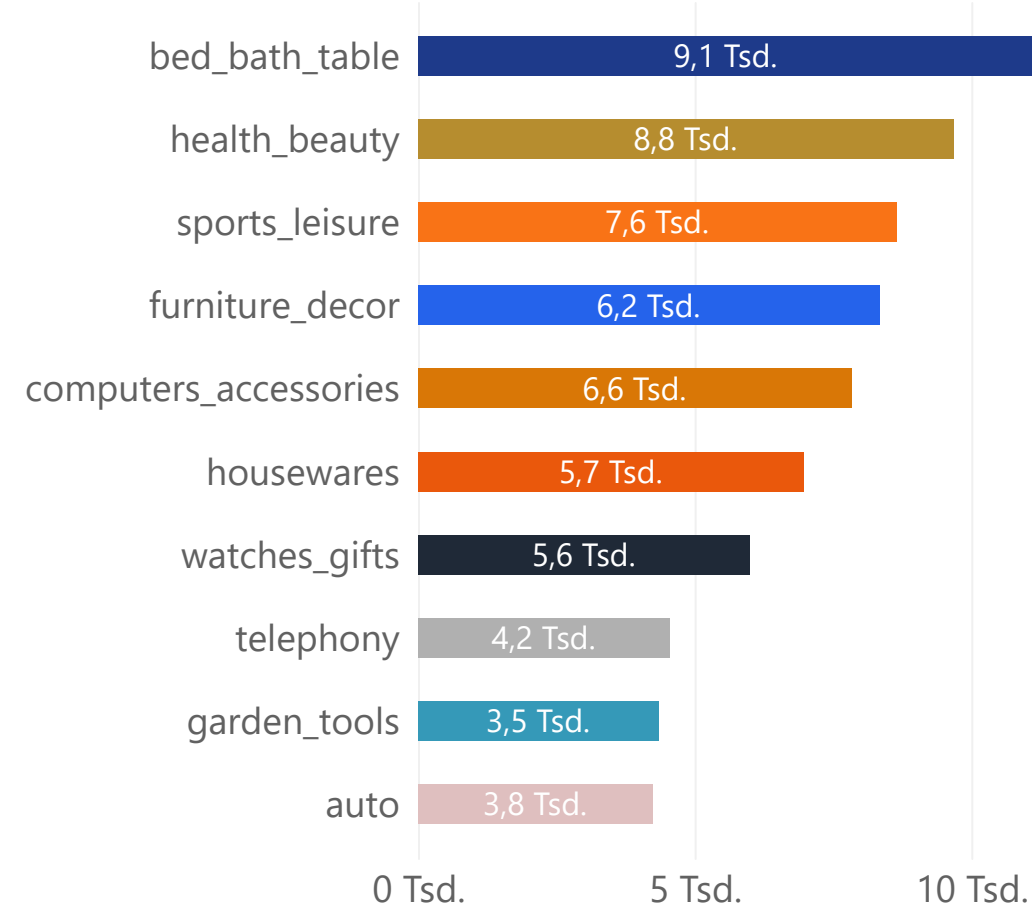
Do most orders come in when the seller isn't working (at the weekend), and does this therefore lead to longer processing times?

Are certain categories responsible for longer processing times?

Weekday of Purchase

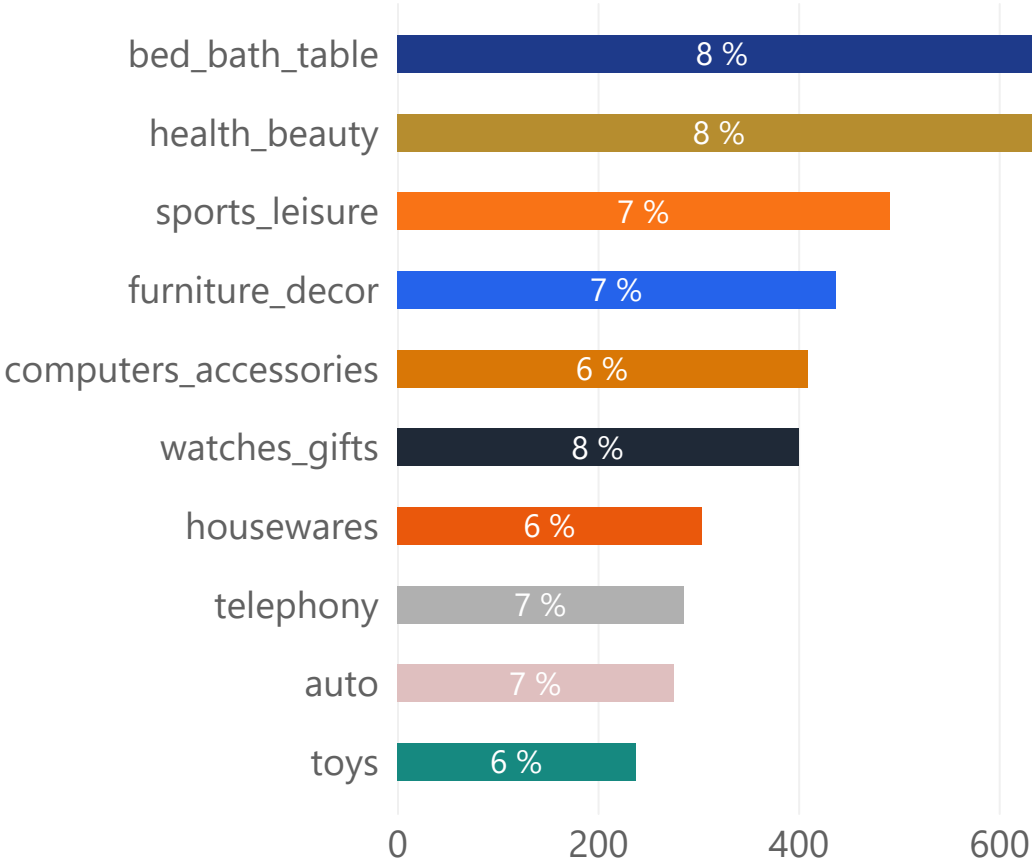


Sold Items per Category, with Number of Total Orders



Flop 10 Categories of delayed deliveries, with delayed Rate

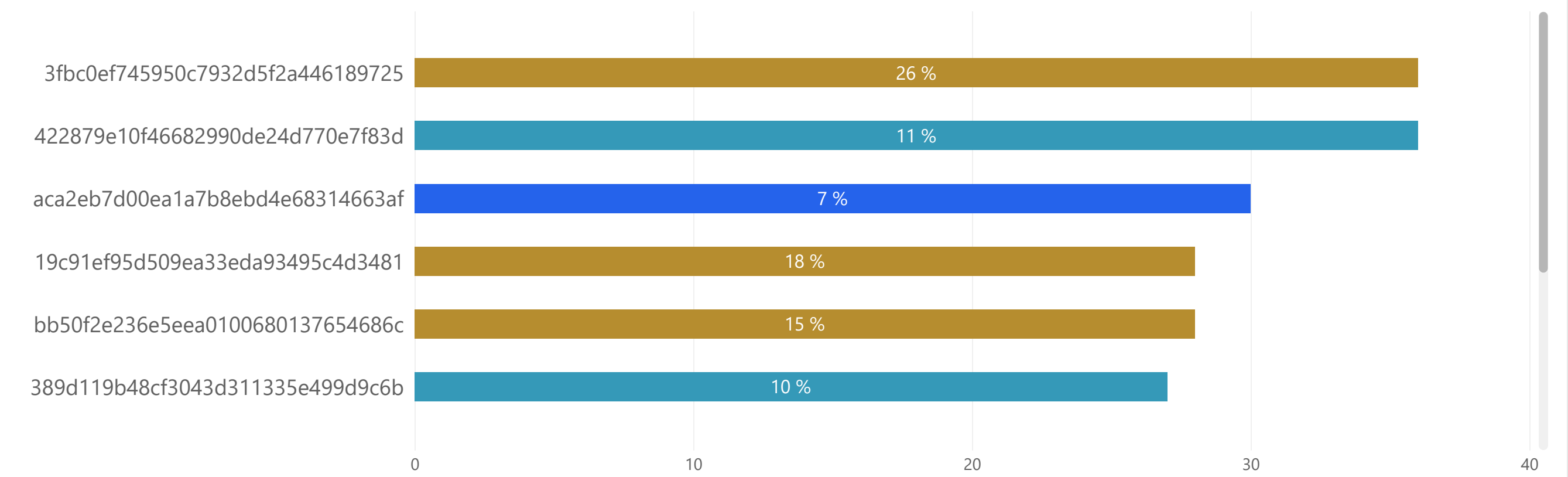
Single Product Orders



Possible Cause: Product Availability

Flop 10 Most Delayed Products, with delayed rate

Single Product Orders



Orders

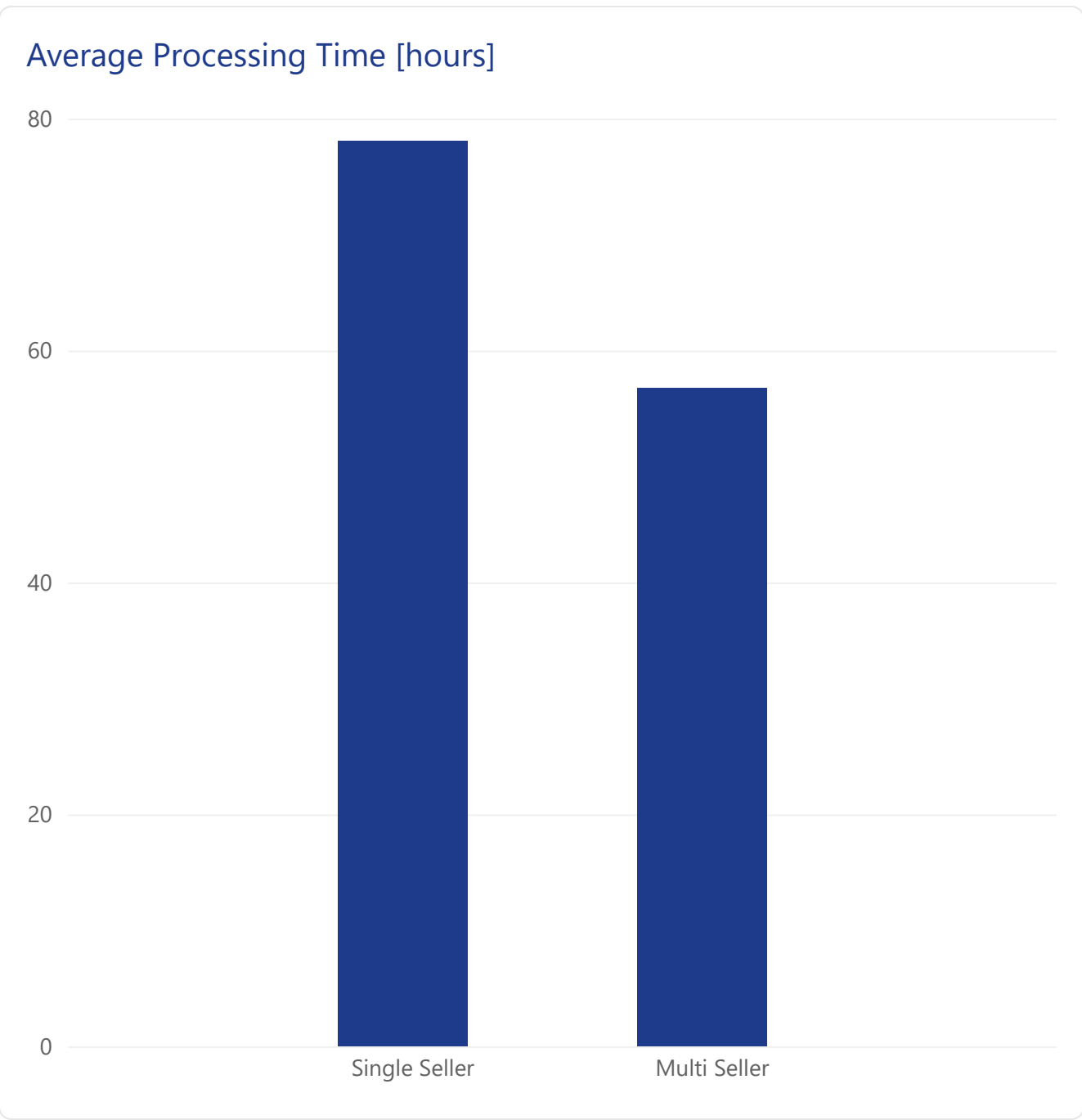
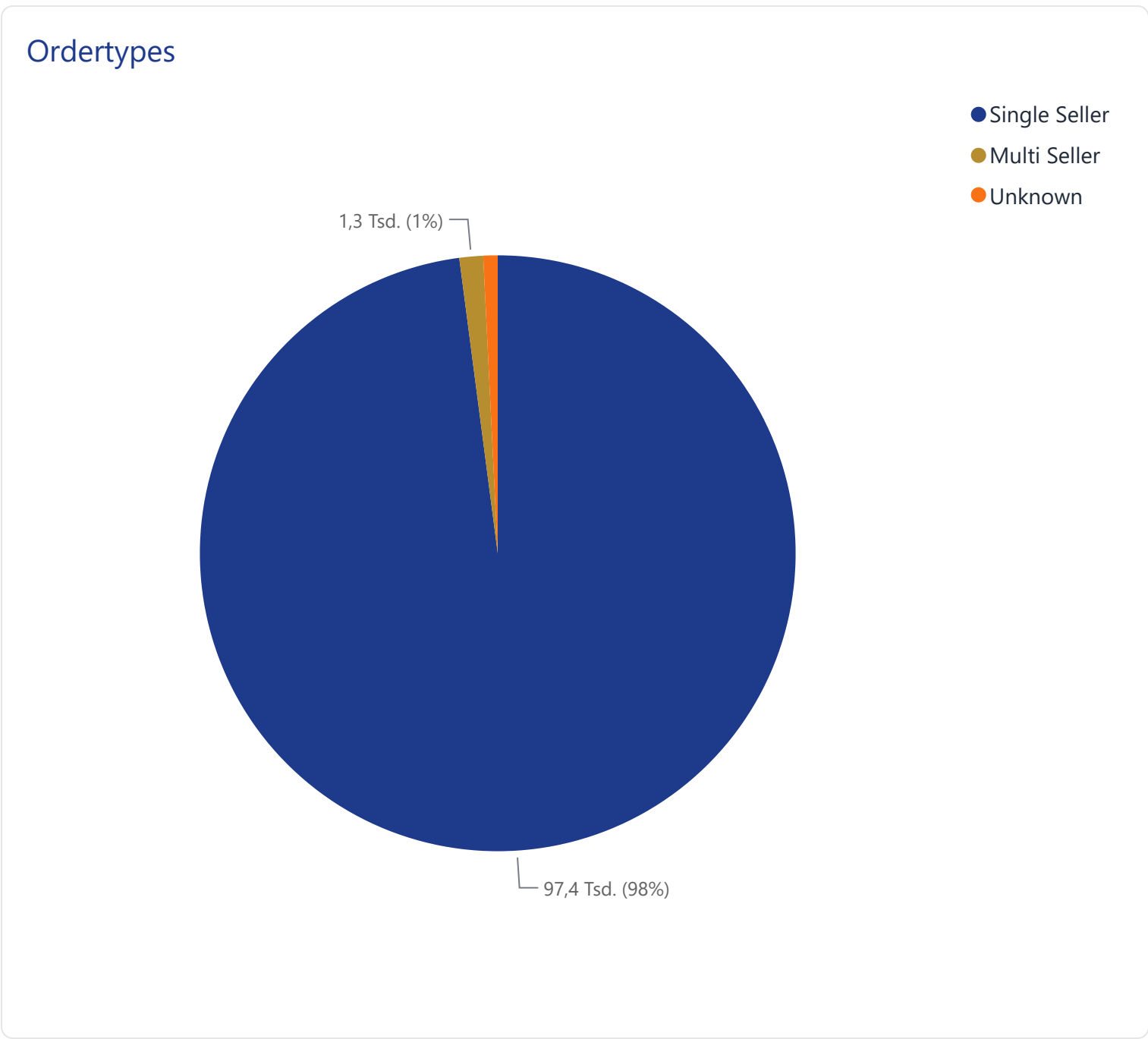
Single Seller Orders

Order ID	Purchase Date	Weekday	Handover to Carrier	Weekday	Processing Time in hours	in days	Delayed Days
da81fbc27b55e0f3d2813cf2078dc780	14.11.2017 21:07:55	Tuesday	20.03.2018 15:44:40	Tuesday	3018	126	100
97f48024fcc76f1898e397ad6966e3a0	29.11.2017 12:25:00	Wednesday	16.03.2018 13:58:02	Friday	2569	107	80
8b7fd198ad184563c231653673e75a7f	14.11.2017 10:04:27	Tuesday	26.02.2018 17:27:15	Monday	2503	104	91
866314550f6d7a55c82917d9b4463e1f	16.11.2017 14:55:04	Thursday	21.01.2018 16:12:17	Sunday	1586	66	43
5cc475c7c03290048eb2e742cd64cb5e	04.10.2016 21:54:52	Tuesday	06.12.2016 17:24:00	Tuesday	1508	63	18
a4a57f1ffa25b90dea9f150fee89db84	06.09.2017 17:07:29	Wednesday	06.11.2017 20:51:56	Monday	1467	61	35
2805499c211b52dfc1e64a1349ef45e2	19.02.2018 15:11:50	Monday	16.04.2018 13:54:50	Monday	1342	56	31
2631dba338efbcea9c3ace77ce210c01	06.10.2016 08:28:19	Thursday	30.11.2016 10:53:48	Wednesday	1322	55	-25
7d86c4aa9e59504b23f16c7ca68954a7	25.01.2017 16:38:10	Wednesday	21.03.2017 14:40:51	Tuesday	1318	55	27
5d6e9993ecc20a59e637ce711858d081	19.01.2017 11:26:57	Thursday	13.03.2017 09:16:51	Monday	1270	53	24

Possible Cause: Large Orders processed by only one Seller

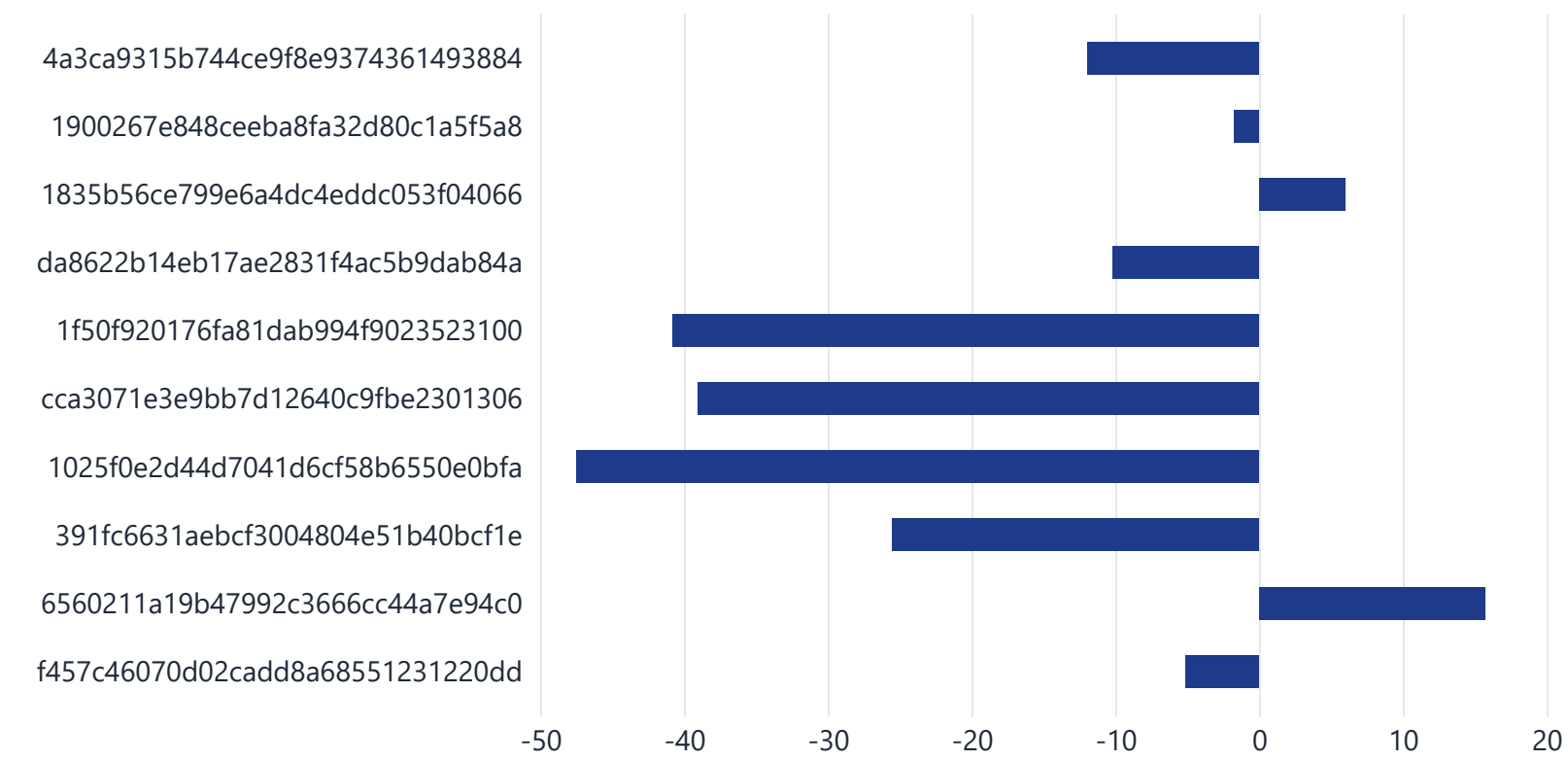
-

Order Splitting as a chance to increase processing performance?



Multi Seller Orders decrease Seller's Processing Time

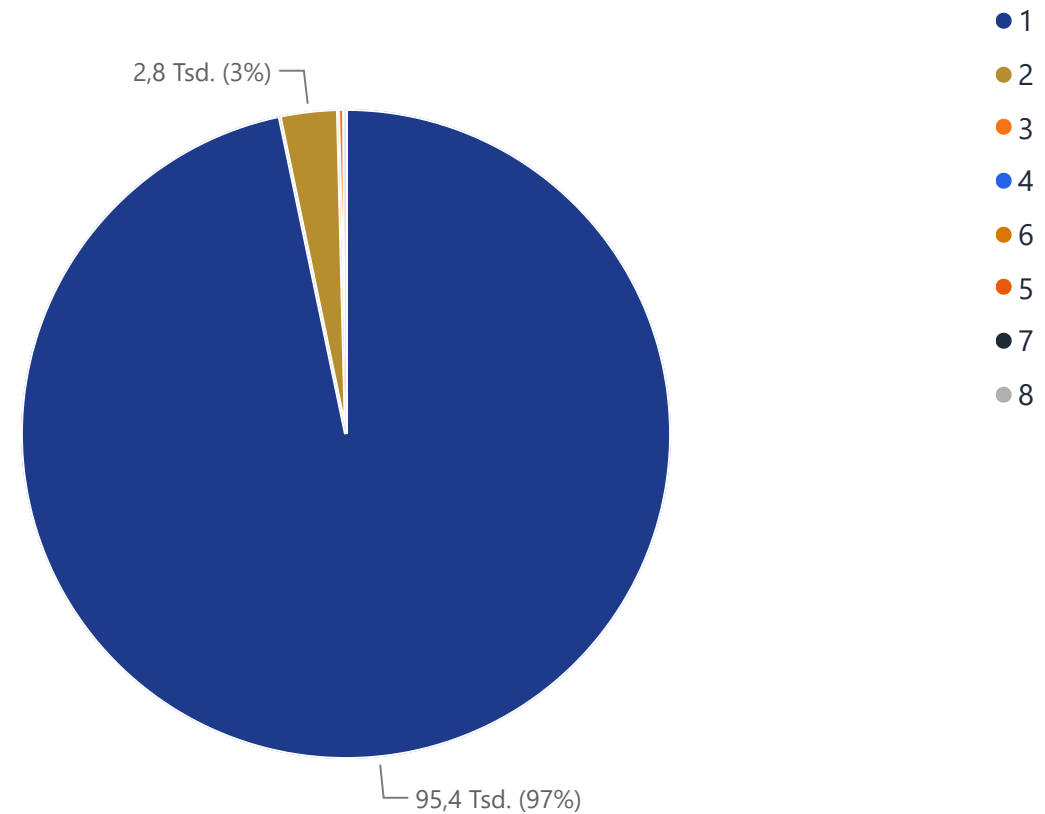
Top 10 Seller, sort by Number of processed Multi Seller Orders descending (AVG Multi Seller Orders - AVG Single Seller Orders)



Splitted Orders increase Seller's Processing Time.

Are there Multi Product Orders to split?

Percentage of Orders containing multiple Products



Seller Performance Analyse

Seller Performance Table

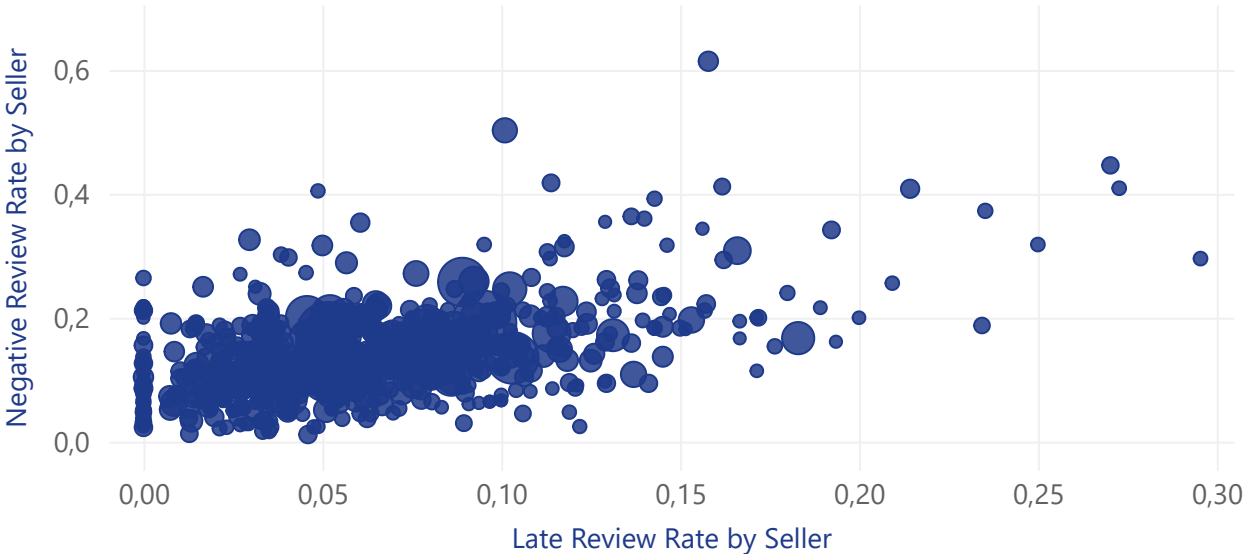
seller_id	Average Review Score by Seller	Review Count by Seller	Negative Review Rate by Seller	Average Delivery Variance by Seller - Reviewed Orders
1ca7077d890b907f89be8c954a02686a	2,33	114	61,40 %	-6,73
2eb70248d66e0e3ef83659f71b244378	2,69	199	50,25 %	-10,10
54965bbe3e4f07ae045b90b0b8541f52	3,00	74	44,59 %	-3,74
a49928bcd77c55c6d6e05e09a9b4ca5	3,01	98	40,82 %	-4,45
972d0f9cf61b499a4812cf0bfa3ad3c4	3,03	79	41,77 %	-9,67
ad781527c93d00d89a11eec9dcad7c1	3,07	44	40,91 %	-9,08
Gesamt	4,10	82195	14,10 %	-11,89

Year

- ☐ 2017
- ☐ 2018

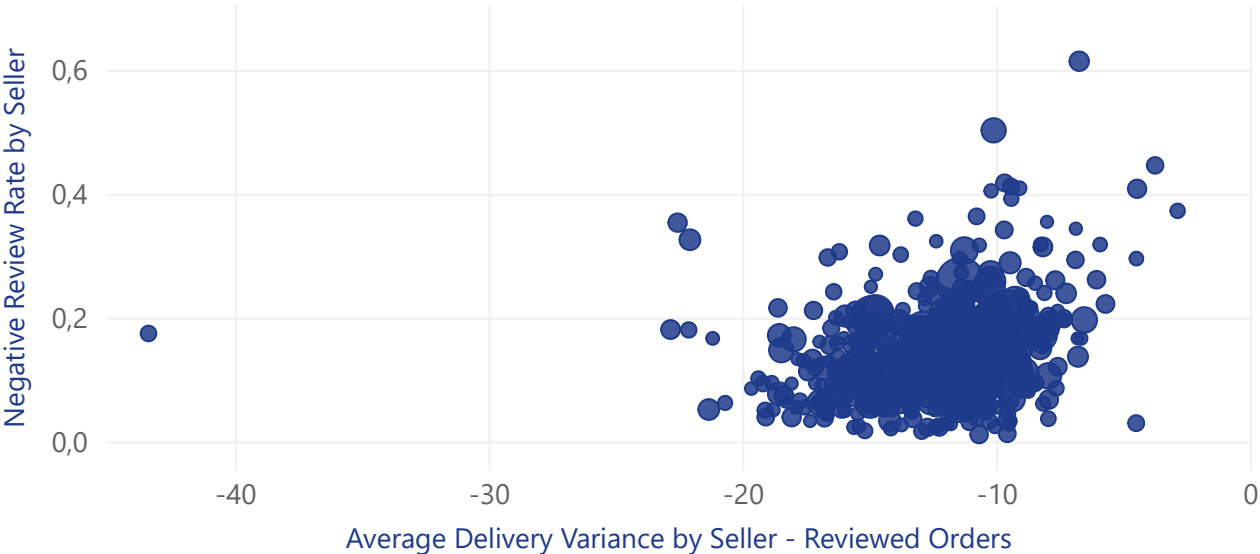
Late Delivery Rate vs. Negative Review Rate

Seller-level relationship, bubble size = review count



Delivery Variance vs. Negative Review Rate

Seller-level relationship, sellers with 30+ reviews



Key Findings

How does seller performance vary in terms of customer satisfaction and delivery performance?

1. Seller performance varies substantially.

Negative review rates differ considerably across sellers, indicating heterogeneous customer experiences.

2. Delivery performance shows an association with negative reviews.

Sellers with higher late-delivery rates tend to show higher negative review rates, although substantial variation remains.

3. Seller-level delivery variance alone does not fully explain customer satisfaction.

The relationship between delivery variance and negative review rate is relatively weak and widely dispersed.

Methodological note:

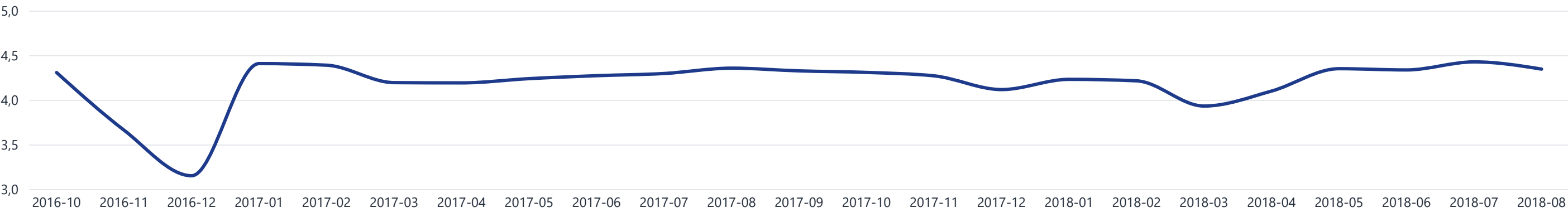
Seller-level relationships indicate associations and do not establish causality. Orders may contain multiple sellers, which limits seller-level attribution.

Customer Satisfaction Analysis

Total Reviews	Average Review Score	Positive Review Rate	Negative Review Rate	Year
99,22 Tsd.	4,09	77,07%	14,69%	2017 2018

Average Review Score Over Time

Monthly development of customer satisfaction



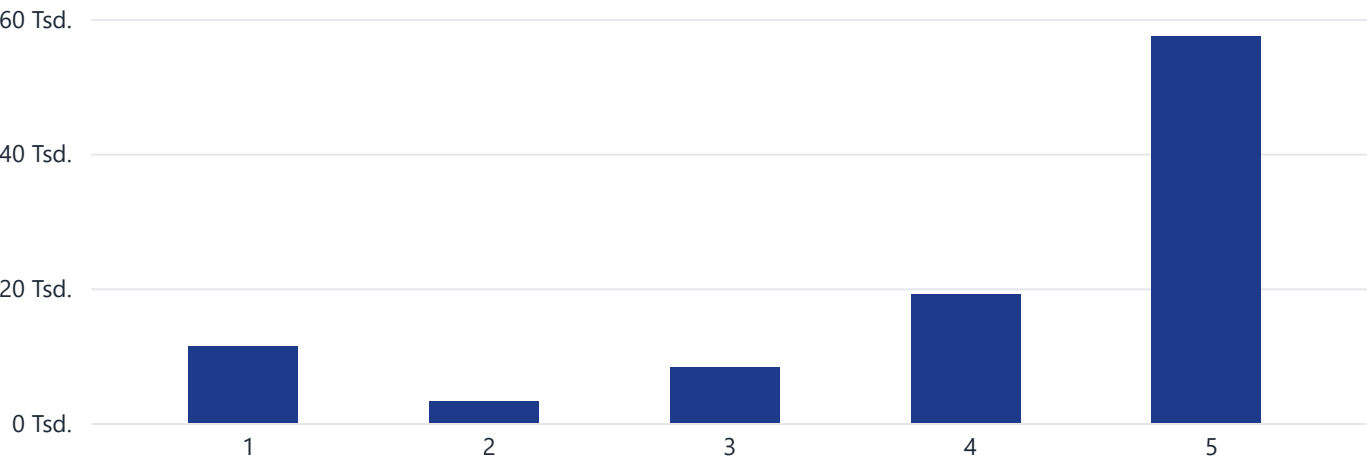
Reviews by Sentiment Category

Distribution of customer reviews



Review Score Distribution

Number of reviews by rating



Key findings

How satisfied are Olist customers and how is customer satisfaction distributed across review ratings and review categories?

1. Overall customer satisfaction is high

Customers give an average review score of **4.09 out of 5**, indicating generally positive customer satisfaction.

2. Positive reviews dominate the review base

77.07% of reviews are classified as positive, while **14.69% are classified as negative**.

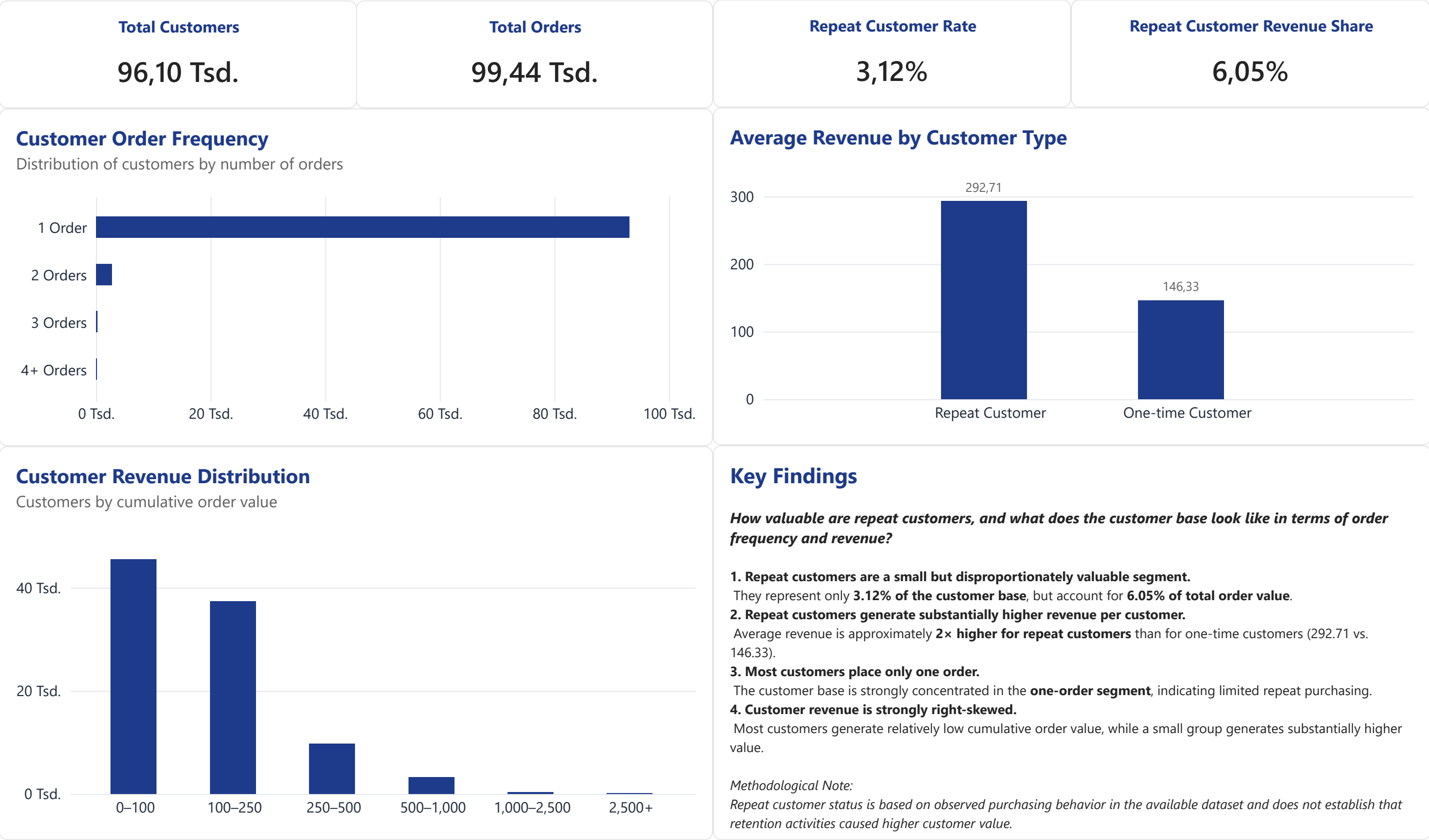
3. 5-star reviews account for the largest share of all ratings

The review distribution is strongly concentrated on **5-star ratings**, while low ratings represent a substantially smaller share.

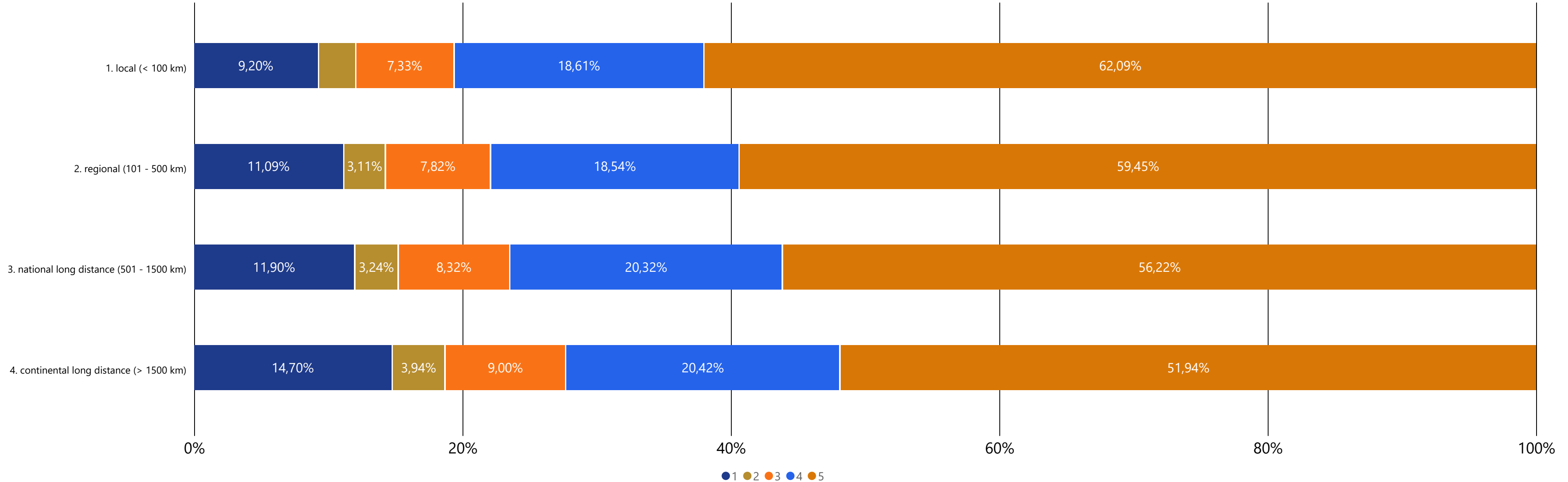
4. Negative feedback represents a relevant minority

Despite the high overall satisfaction, almost **15% of reviews are classified as negative**, indicating a meaningful group of dissatisfied customers.

Customer Analytics - Loyalty & Customer Value



how distance impacts reviews



Key findings:

- The reviewscore decreases with rising distance
- The one star count increases in continental distance by around 3 %, while the 5 star reviews, decrease by around 4 %
- The change of distances from local to regional, and from national long distance to continental long distance, has the highest decrease of customer satisfaction

Delivery Performance Analysis

Reviews on Late Orders

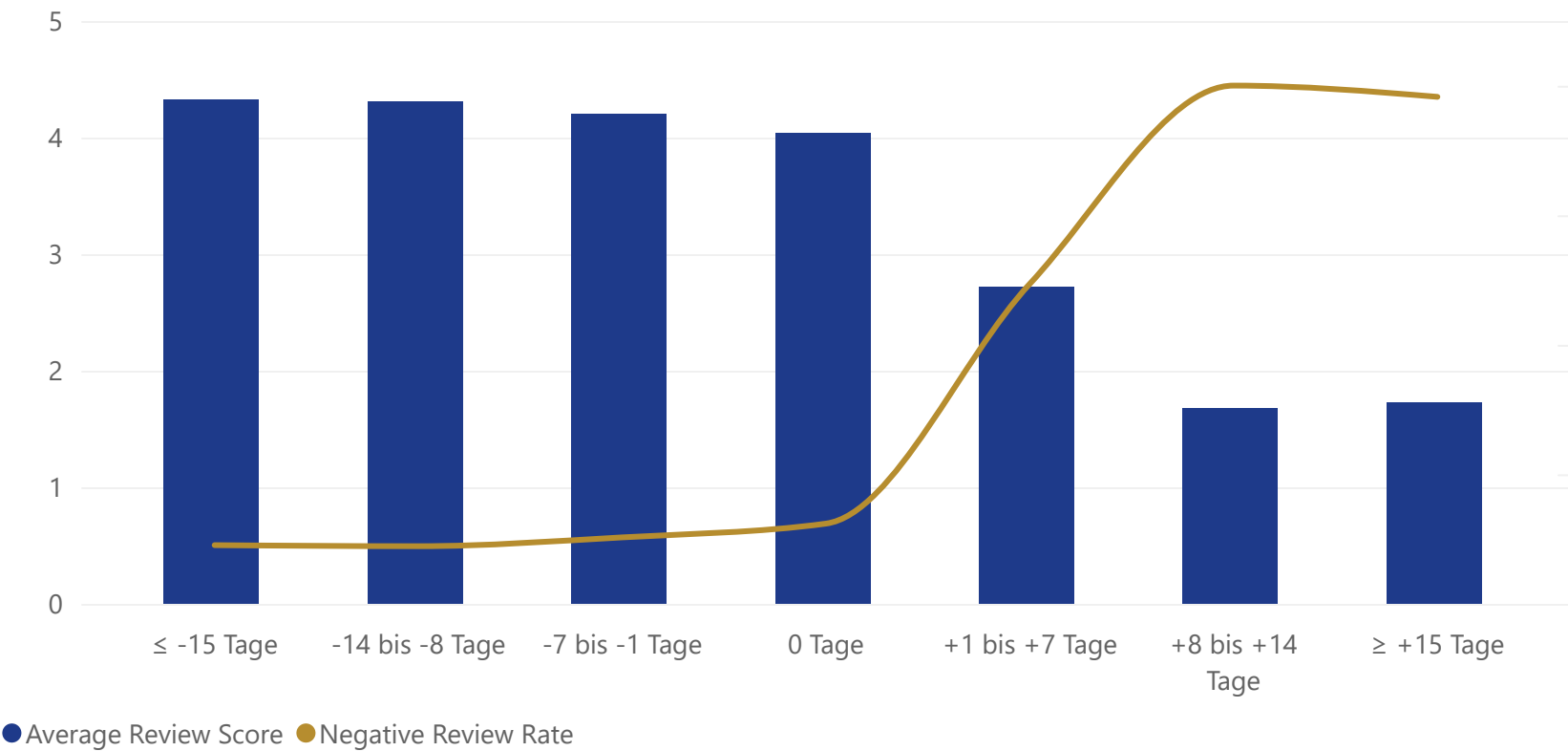
6.409

Late Review Rate

6,46%

Customer Satisfaction by Delivery Performance

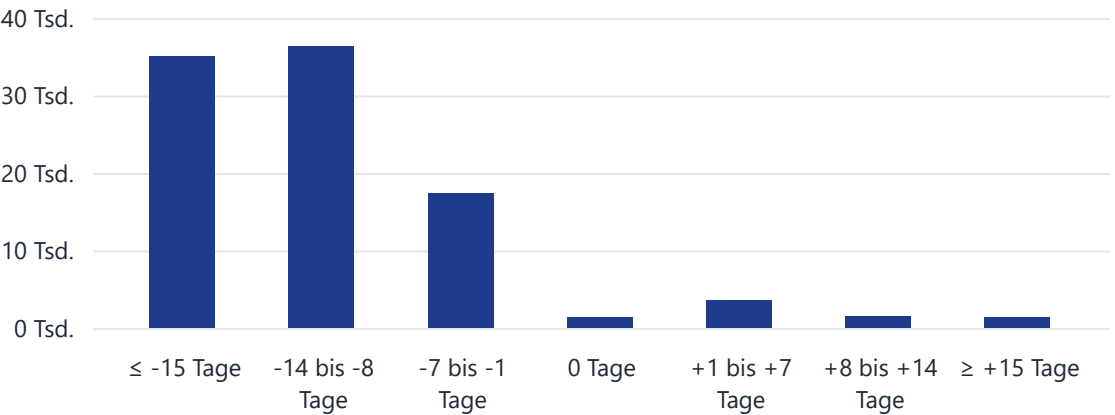
Average review score and negative review rate



● Average Review Score ● Negative Review Rate

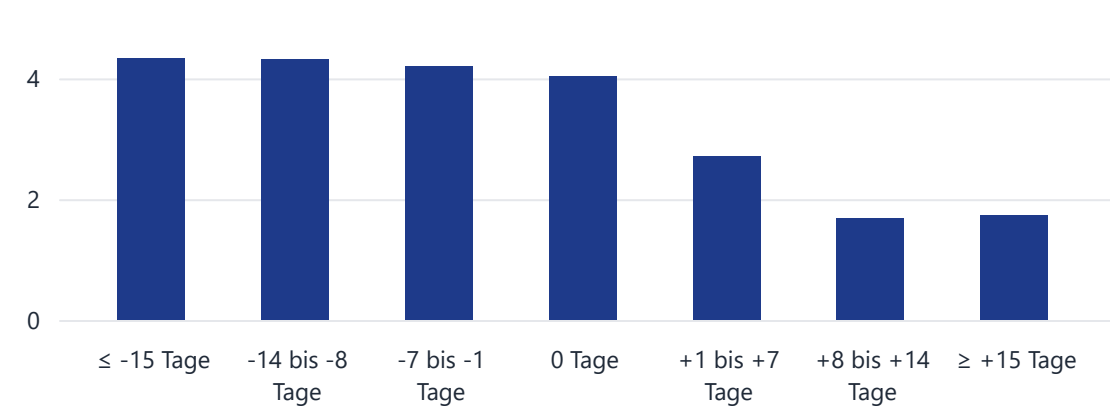
Reviews by Delivery Variance

Number of reviews by delivery variance bucket



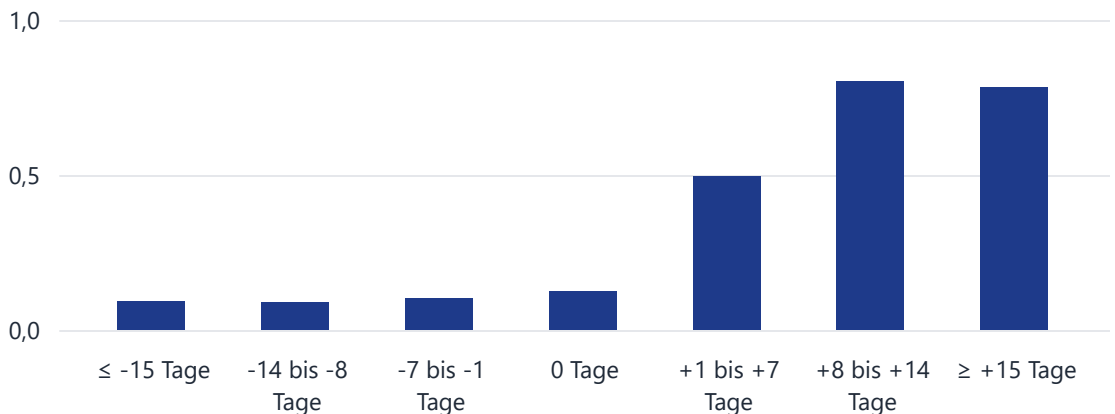
Average Review Score by Delivery Variance

Average rating by delivery variance bucket



Negative Review Rate by Delivery Variance

Share of negative reviews by delivery variance bucket



Year

○ 2017

○ 2018

Key Findings

How does delivery performance affect customer satisfaction and negative reviews?

1. Delivery delays are strongly associated with lower customer satisfaction.

Average review scores remain above 4.0 for orders delivered on or before the estimated date, but decline substantially for orders delivered more than one week late.

2. Negative reviews increase sharply with delivery delays.

The negative review rate rises strongly once orders are delivered more than 7 days after the estimated delivery date.

3. Most reviewed orders are delivered ahead of the estimated date.

The largest review volumes are concentrated in orders delivered at least one week before the estimated delivery date.

Methodological Note

The analysis shows an association between delivery variance and customer satisfaction; it does not establish causality.

Product Category Analysis

Total Reviews

98,47 Tsd.

Negative Review Rate

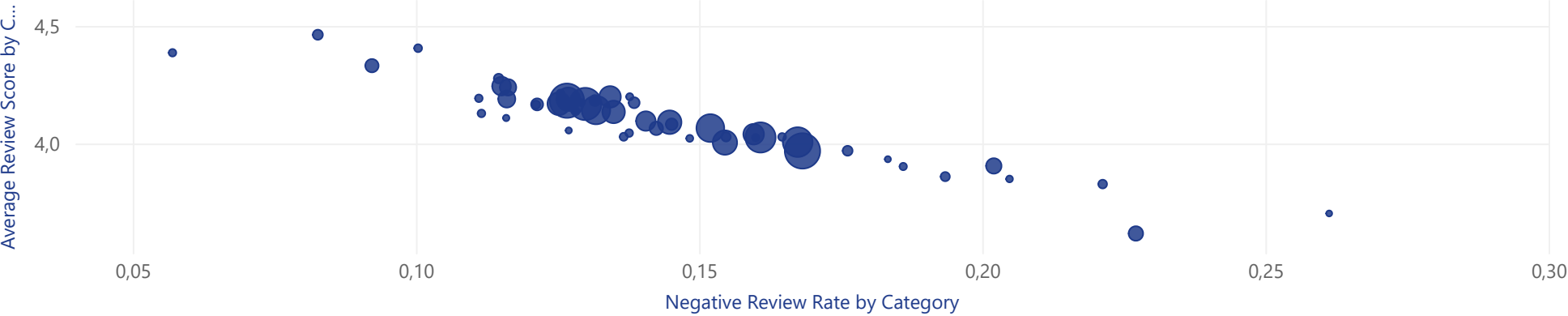
14,19%

Avg. Review Score

4,10

Higher Negative Feedback Is Associated with Lower Ratings

Bubble size represents review volume



Category Performance Details

product_category_name	Negative Review Rate by Category	Average Review Score by Category	Review Count by Category
fashion_male_clothing	26,13 %	3,70	111
office_furniture	22,71 %	3,62	1268
audio	22,13 %	3,83	348
construction_tools_safety	20,48 %	3,85	166
home_comfort	20,20 %	3,90	1470
Gesamt	14,20 %	4,10	97652

Key Findings

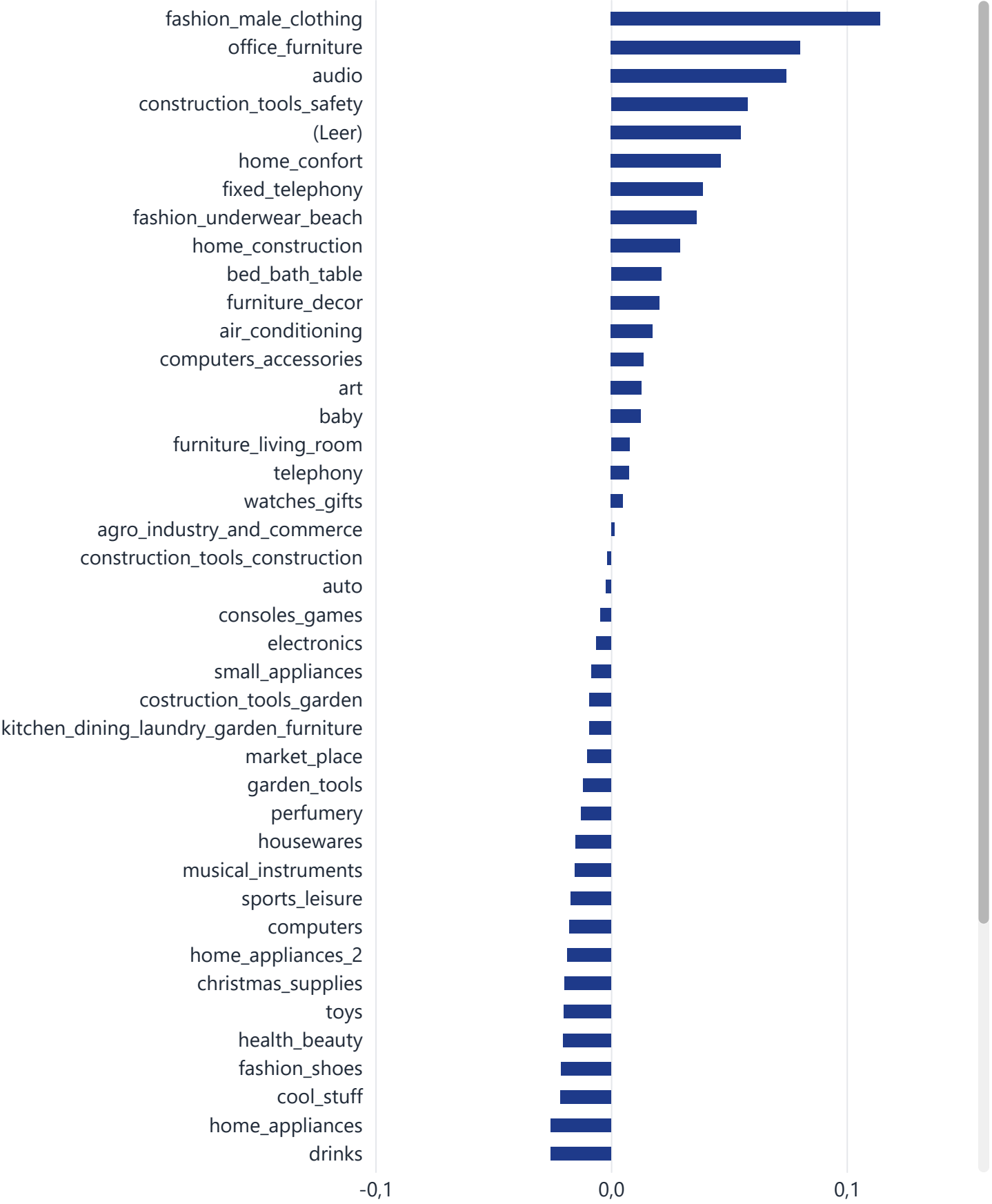
Which product categories generate the most negative customer feedback, and how does satisfaction vary across categories?

- 1. Negative review rates vary substantially across product categories.**
Some categories perform considerably above the marketplace average, indicating category-specific customer satisfaction challenges.
- 2. Higher negative review rates are associated with lower average review scores.**
The category-level relationship shows a clear negative pattern between negative review rate and average review score.
- 3. Review volume differs strongly across categories.**
The distribution of review volume indicates that customer feedback is concentrated in a limited number of categories.

Methodological note:
Category-level relationships indicate associations and do not establish causality. Categories with low review volumes should be interpreted with caution.

Category Negative Review Rate vs. Marketplace

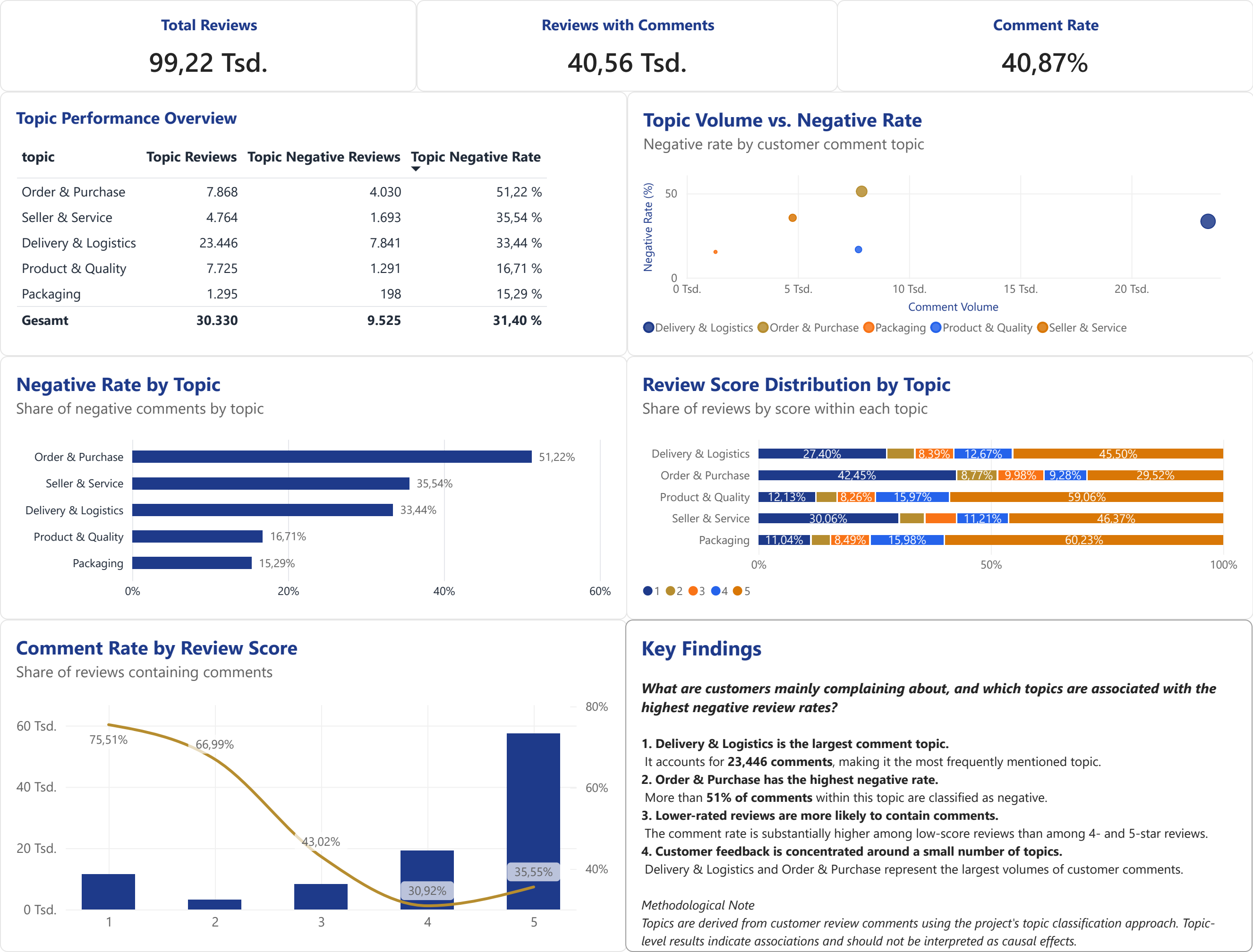
Difference from overall negative review rate



Year

- ☐ 2017
- ☐ 2018

Comment Analysis - Review Text Analysis



Predicting Top Customer Reviews (4+ Stars)

Driven by Delivery Distance, Processing Time, Delivery Time, and Delays.

Logistics and processing is responsible for 40 % of negative reviews

Accuracy: 75 %

Cross-Val F1-Score: 75.08 %

Evaluationmatrix

Results

- Accuracy: 75.27%
- Cross-Val F1-Score: 75.08%

Precision & Recall:

	precision	recall	f1-score	support
Schlecht	0.41	0.40	0.40	4062
Gut	0.84	0.85	0.84	15210
accuracy			0.75	19272
macro avg	0.63	0.62	0.62	19272
weighted avg	0.75	0.75	0.75	19272

Confusionmatrix

